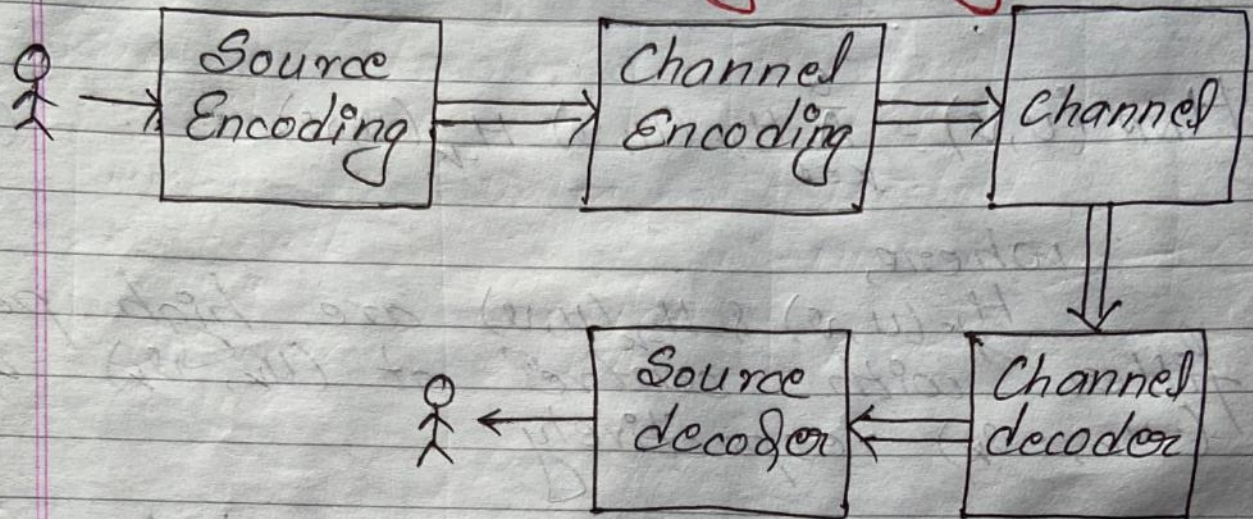


Introduction to Image Coding & Compression



(fig: Block diagram of Image Compression)

Image Compression addresses the problem of reducing the amount of data required to represent a digital image.

The underlying basis of the reduction process is removal of redundant data.

The compressed image is decompressed to reconstruct the original image and approximation of it.

Image (data) compression is the reduction of number of bits required to transmit the image.

(2)

probably without any remarkable loss of information.

The image encoding process through the channel is called image compression and image decoding process is known as image decompression.

Application of Data Compression

- (i) Broadcasting TV
- (ii) Remote Sensing through satellite
- (iii) Military Communication
- (iv) Facsimile transmission & telecommunication (Fax)

gmf

Data Redundancy:

It contains data that either provide no relevant information or simply restate that which is already known. This is called data redundancy.

Image Compression Technique:

- (i) Pixel Coding Technique
- (ii) Predictive Coding
- (iii) Transform Coding
- (iv) Other methods

(3)

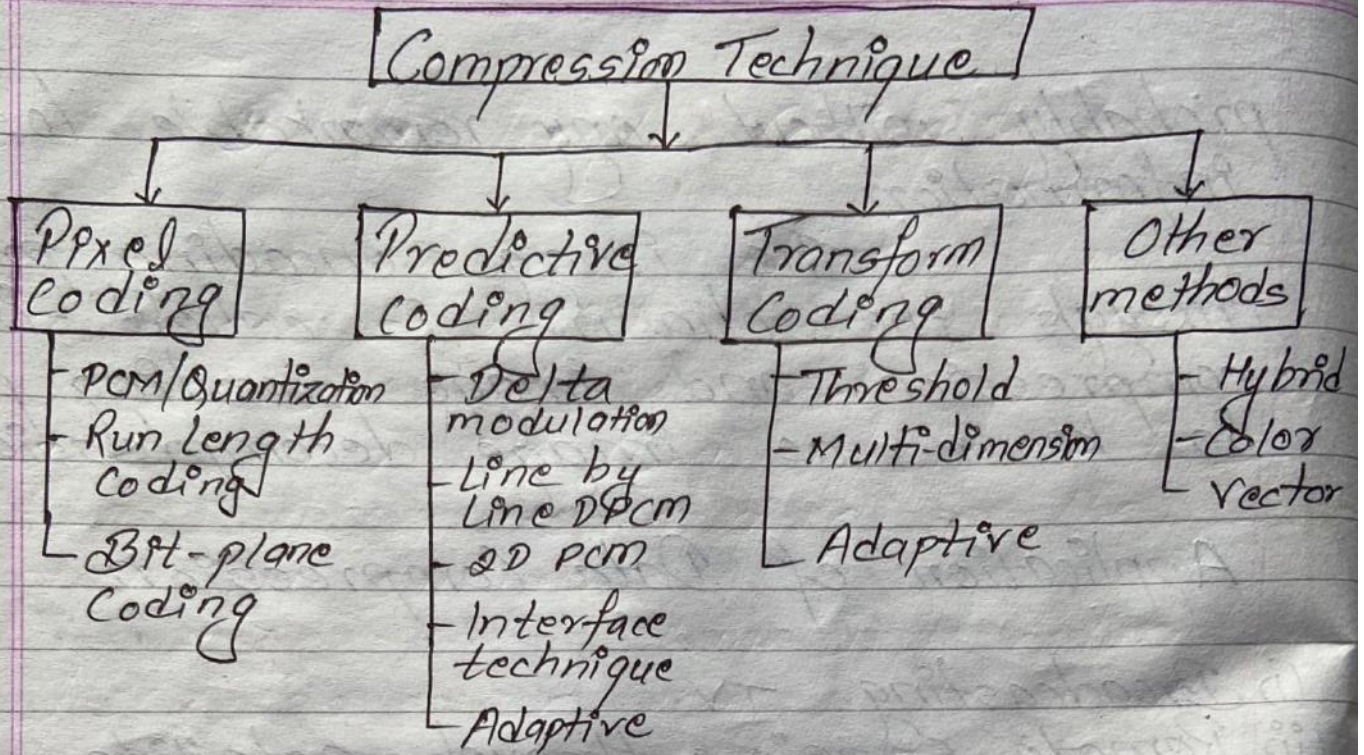


fig: Different Image Compression Technique

Q.1) Pixel Coding Technique

In this technique, each pixel is processed independently ignoring the inter-pixel dependencies.

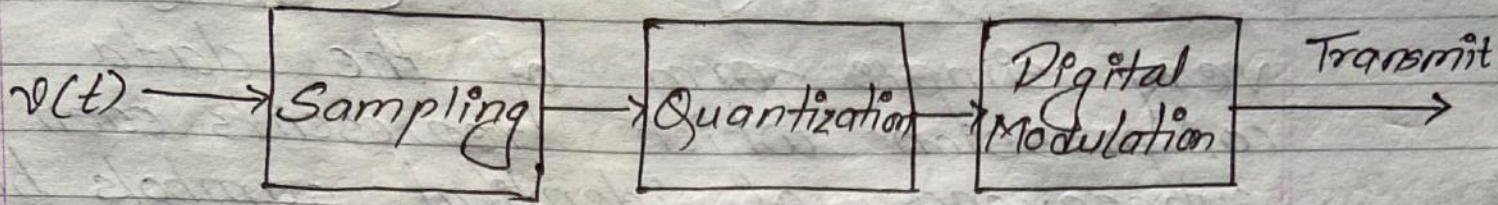
This technique firstly remove the inter dependencies between the neighbourhood pixel and then removal of redundant data.

Pixel Coding Technique is further divided as:

- (a) PCM / Quantization
- (b) Run length Coding
- (c) Bit-plane Coding

(4)

A) Pulse Code Modulation (PCM):



(fig: PCM / Quantization)

In PCM technique, the incoming video signal is sampled, quantized and coded by the suitable code word before inputting it to a digital modulator for transmission.

The quantized output is coded by a fixed length binary code word having usually 8-bits.

The formula for achieving rate of compression of PCM is given by the rate distortion formula.

$$\text{i.e. } R_{\text{PCM}} = \frac{1}{2} \log_2 \left(\frac{\sigma_v^2}{\sigma_q^2} \right), \quad \sigma_q^2 \leq \sigma_v^2$$

where,

σ_v^2 = variance of quantizer i/p

σ_q^2 = quantizer mean square deviation

(3)

B) Run length Coding:

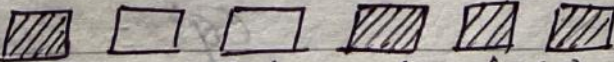
It is a very simple form of the data compression which represents each subsequence of identical symbols by a pair like (L, a) where, L is the subsequence and a is the recurring symbols.

Run length coding describes each row of the image by a sequence of length that describes successive runs.

Example: 'aaobbbbaaa' is coded as '3a3b3a'.

The most common methods are as follows:

- (i) To specify the value of the first run of each row
- (ii) To assume that each row begins with a white run

for example: 

black(1) white(2) black(3)

(6)

from above figure, the run length code can be written as:
1 black 2 white 3 black

In this technique, the numeric run length coding can also be defined as:
00000 2222111 = 0/5 2/4 1/3

Entropy Coding:

Entropy codes encoding the given set of symbols with minimum number of bits required to represent them.

The most important entropy coding is Huffman coding. The theoretical minimum average of bits that are required to transmit a particular source string is known as entropy of source & it can be computed by using the following formula:

$$\text{Entropy (H)} = - \sum_{i=1}^N P_i \log_2 P_i$$

It has two major parts:
(i) Construction of probability tree
(ii) Assigning code to each node of the constructed tree

(7)

It gives the variable length code words and highest probability assigns ~~sub~~ ~~for~~ short path and lowest probability assign longest path i.e. ~~the~~ code length.

Huffman Coding Algorithm

- (i) Arrange the symbol probabilities P in an decreasing order & consider them as leaf node of a tree.
- (ii) While there is more than one node,
 - (a) Merge the 2 nodes with smallest probability to form a new node.
 - (b) Assign 1 and 0 to each pair of branches merging into a node.
- (iii) Read sequentially from root node to leaf node.

Q Source generates the symbol S_1, S_2, S_3, S_4 and S_5 randomly with probability $P_1 = 0.4, P_2 = 0.2, P_3 = 0.2, P_4 = 0.1$ & $P_5 = 0.1$ respectively.

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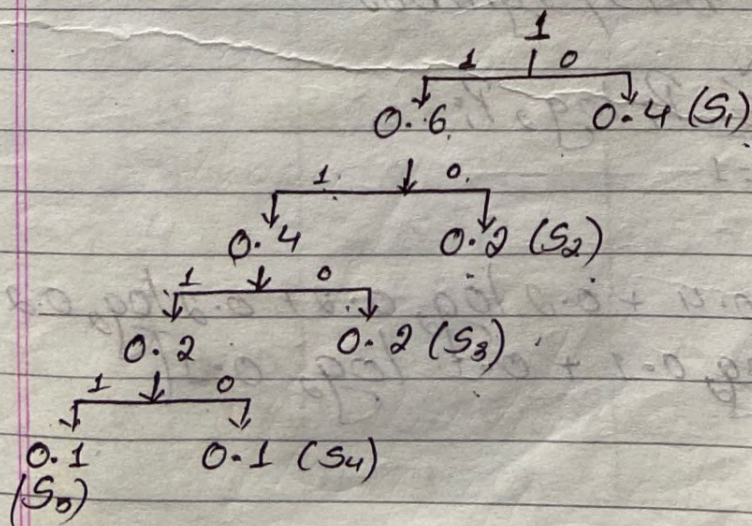
Generate the codeword for each symbol using Huffman coding. Also calculate the compression ratio & efficiency of the system.

Solⁿ Step 1:

Symbol	Original Size	Probability (P_i)	step 1	step 2	3	4
S_1		0.4	0.4	0.4	0.6	1
S_2		0.2	0.2	0.4	0.4	
S_3		0.2	0.2	0.2		
S_4		0.1	0.2			
S_5		0.1				

Step 2:

Construct Huffman tree, assume highest probability = 0.6 = 1 & lowest probability 0.4 = 0 (step 3 or 2)



(9)

Step 3

Generate codeword length

<u>Symbol</u>	<u>Probability (P_i)</u>	<u>Code Length (L_i)</u>
S_1	0.4	0 = 1
S_2	0.2	10 = 2
S_3	0.2	110 = 3
S_4	0.1	1110 = 4
S_5	0.1	1111 = 4

Now,

$$\text{Compression ratio (Lavg)} = \sum_{i=1}^N P_i L_i$$

$$= 0.4 \times 1 + 0.2 \times 2 + 0.2 \times 3 + 0.1 \times 4 + 0.1 \times 4$$

$$= 0.4 + 0.4 + 0.6 + 0.4 + 0.4$$

$$= 2.2 \text{ bits/symbol}$$

$$\text{Entropy (H)} = - \sum_{i=1}^N P_i \log_2 P_i$$

$$= -[0.4 \log_2 0.4 + 0.2 \log_2 0.2 + 0.2 \log_2 0.2 + 0.1 \log_2 0.1 + 0.1 \log_2 0.1]$$

$$= 2.12$$

$$\text{Efficiency} = \frac{\text{Entropy}}{\text{Larg}} \times 100\%$$

$$= \frac{2.12}{2.2} \times 100\%$$

$$= 96.36\%$$

Q. Calculate the compression ratio and efficiency from the image given below:

r	0	1	2	3	4	5	6	7
n_k	400	1350	659	2034	816	2560	250	1500

Solⁿ

Step 1: Calculate the probability in each gray level.

Gray level	no. of pixels	Probability (P_i) = n_k/n
0	400	0.041
1	1350	0.141
2	659	0.068
3	2034	0.212
4	816	0.085
5	2560	0.267
6	250	0.026
7	1500	0.156
$n = 9569$		

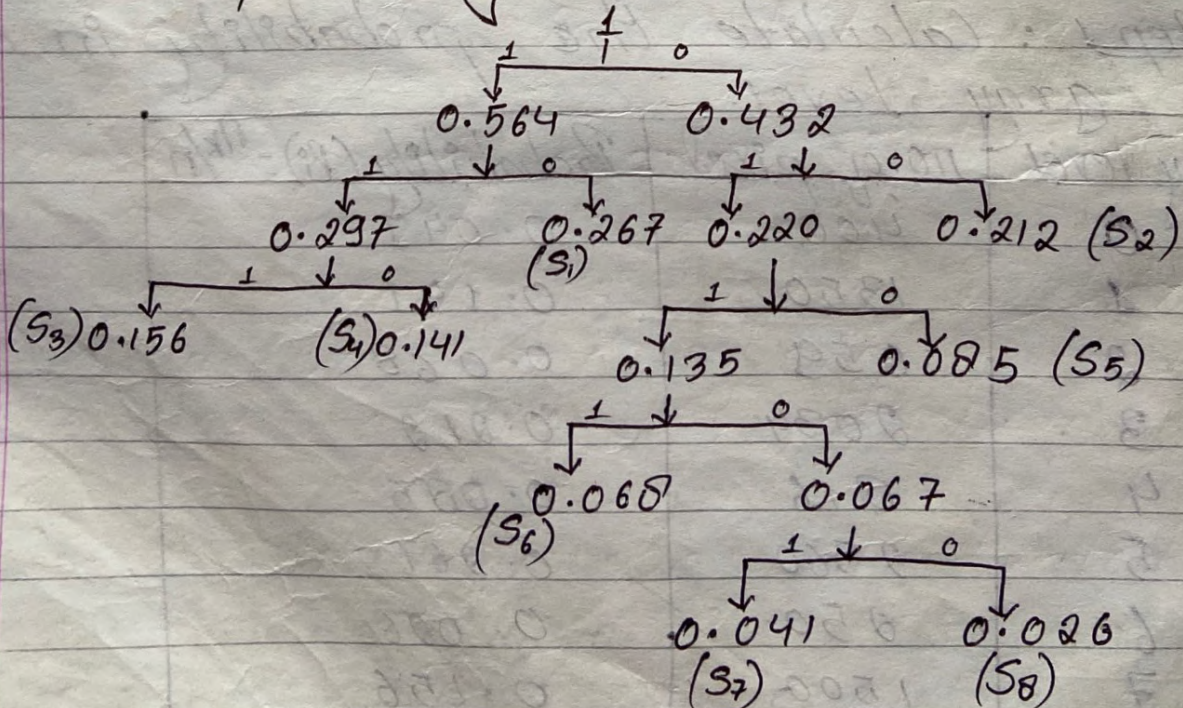
(10)

Step 2:

Assume symbol for each probability in descending order

Symbol	P_i	step						
		1	2	3	4	5	6	7
S_1	0.267	0.267	0.267	0.267	0.297	0.432	0.564	0.99
S_2	0.212	0.212	0.212	0.220	0.267	0.297	0.432	
S_3	0.156	0.156	0.156	0.212	0.220	0.267		
S_4	0.141	0.141	0.141	0.156	0.212			
S_5	0.085	0.085	0.135	0.141				
S_6	0.068	0.068	0.085					
S_7	0.041	0.067						
S_8	0.026							

Step 3: Construct Huffman tree, assume highest probability = 1 & lowest = 0



Step 4 : Generate Codeword length

<u>Symbol</u>	<u>Probability (P)</u>	<u>Code length (L_i)</u>
S ₁	0.267	10 = 2
S ₂	0.212	00 = 2
S ₃	0.156	111 = 3
S ₄	0.141	110 = 3
S ₅	0.085	010 = 3
S ₆	0.068	01101 = 4
S ₇	0.041	01111 = 5
S ₈	0.026	01110 = 5

Now,

$$\text{Compression ratio (Lavg)} = \sum_{i=1}^N P_i L_i$$

$$= 0.267 \times 2 + 0.212 \times 2 + 0.156 \times 3 + 0.141 \times 3 + 0.085 \times 3 + 0.068 \times 4 + 0.041 \times 5 + 0.026 \times 5$$

$$= 0.534 + 0.424 + 0.468 + 0.423 + 0.255 + 0.272 + 0.205 + 0.13$$

$$= 2.711$$

$$\text{Entropy (H)} = - \sum_{i=1}^N P_i \log_2 P_i$$

$$= - \left[0.267 \log_2 (0.267) + 0.212 \log_2 (0.212) + 0.156 \log_2 (0.156) + 0.141 \log_2 (0.141) + 0.085 \log_2 (0.085) + 0.068 \log_2 (0.068) + 0.041 \log_2 (0.041) + 0.026 \log_2 (0.026) \right]$$

$$= 2.687$$

(13)

Now,

$$\text{Efficiency}(\eta) = \frac{\text{Entropy}}{\text{Log}} \times 100\%$$

$$= \frac{2.687}{2.711} \times 100\%$$

$$= 99.11\%$$

c) Bit plane Coding:

It breaks the image into bit planes and apply run length coding to each plane.

Let, 'I' be an image where every pixel value is n-bit long.

We can express every pixel in binary using n-bit from out of an image and 'n' binary matrix (called bit plane) where nth matrix consist of ith bit of the pixel of I.

It can be illustrated as:

$$I = \begin{bmatrix} 101 & 110 \\ 111 & 011 \end{bmatrix}, \text{ corresponding 3 bit plane \& 2x2 matrix.}$$

Now,

$$\bar{I}' = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Bit-plane coding uses the run length coding but in case of m -bit gray level scale image at polynomial 2-base is defined as:

$$a_{m-1} 2^{m-1} + a_{m-2} 2^{m-2} + \dots + a_1 2^1 + a_0 2^0$$

g m

Predictive Coding

Future value will be predicted on the basis of past values. If we know that, the past behaviour of a signal upto a certain point in time then it is impossible to make some prediction about its future values then such process is called prediction. It is of two types.

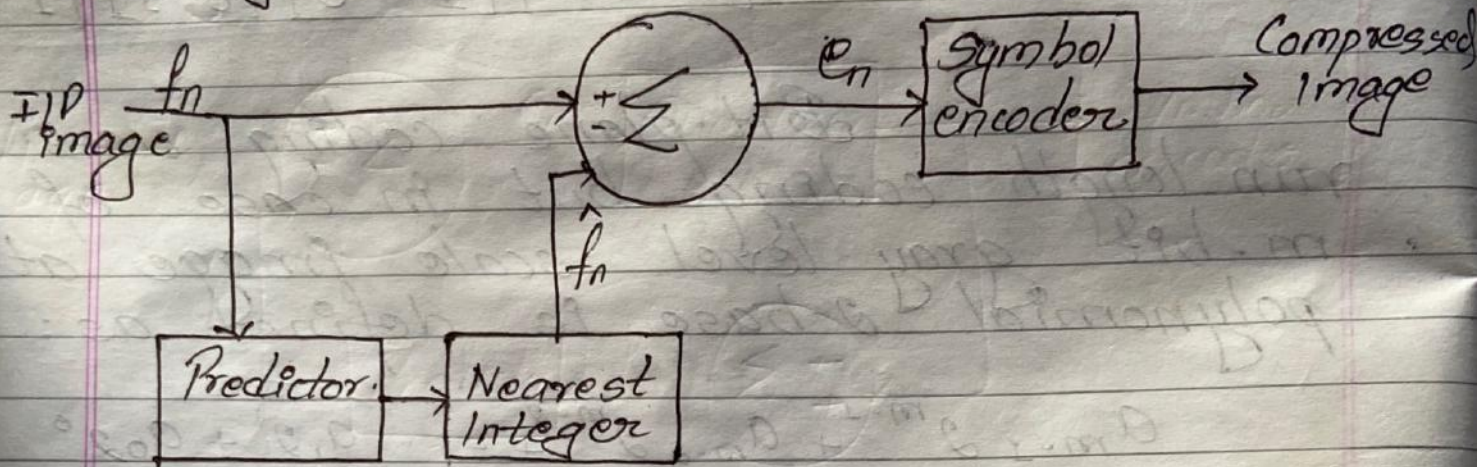
They are:

- (i) Lossless Predictive Coding
- (ii) Lossy Predictive Coding

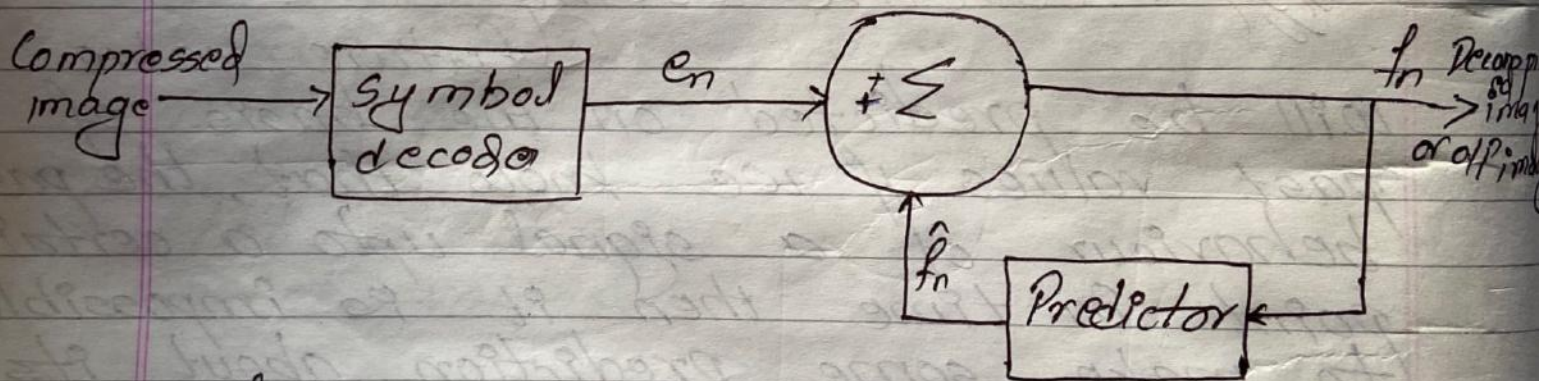
1) Lossless Predictive Coding

There is no information loss & image can be reconstructed exactly the same as the original. So, compression

process is reversible and lossless coding.



(Fig (a) Encoder)



(Fig (b) Decoder)

(Fig: Lossless Predictive Coding)

The system consists of an encoder and a decoder, each containing an identical predictor.

The predictor generates anticipated value of that pixel

based on some number of past values. The o/p is then rounded to the nearest integer.

Then, the differences or prediction error is calculated as:

$$e_n = f_n - \hat{f}_n$$

where,

e_n = prediction error

f_n = i/p image symbol

$\hat{f}_n$ = prediction function.

In case of encoding, symbol encoder uses variable length coding technique to encode symbol.

In case of decoding, symbol decoder uses decoding function as:

$$f_n = e_n + \hat{f}_n$$

where,

f_n = decompressed image or o/p image

In most cases, the predictor is formed by a linear combination of m previous pixel.

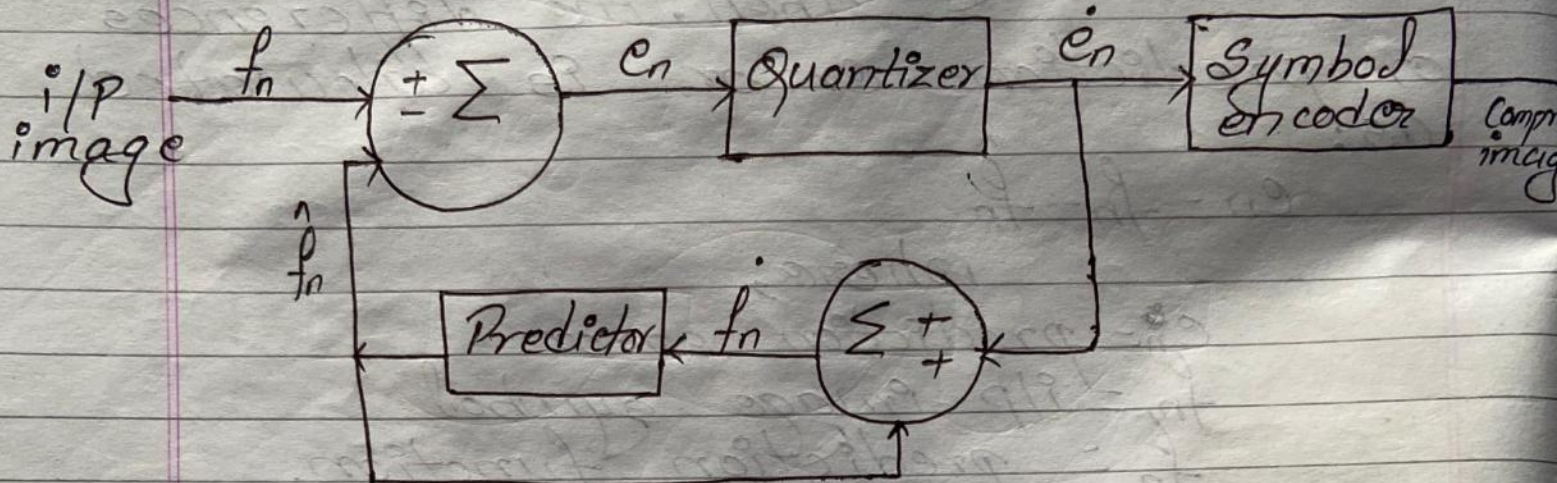
$$\text{i.e. } \hat{f}_n = \text{round} \sum_{i=1}^m \alpha_i f_{n-i}$$

2

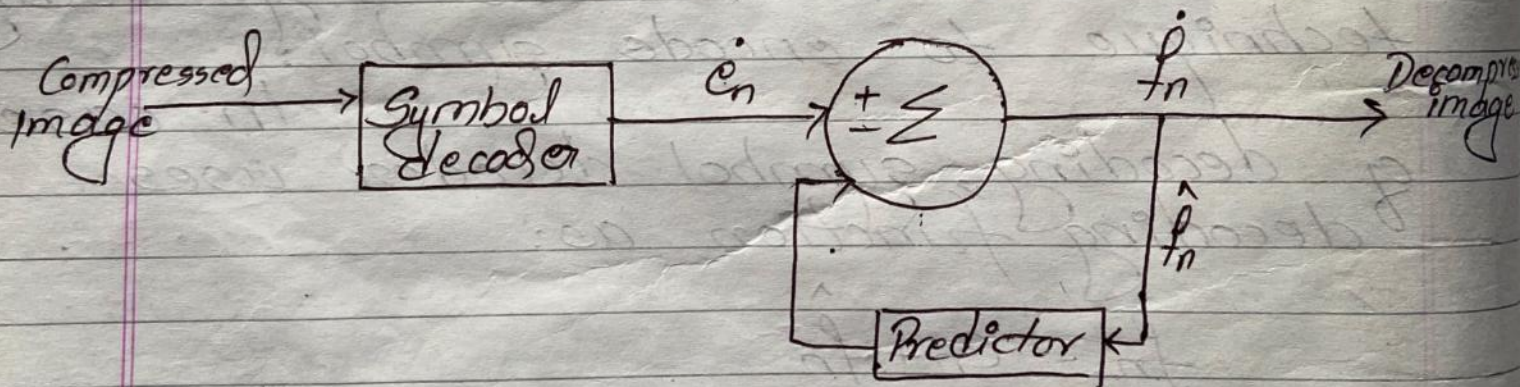
α = predictor coefficient
 m = order of linear prediction

(17)

2) Lossy Predictive Coding



(fig: Encoder)



(fig: decoder)

The major difference in this technique is that a quantizer is added between the symbol encoder and the point where the prediction error is calculated.

The quantizer absorbs the nearest value of a prediction error & maps it into a limited range of output denoted by e_n at encoding and f_n at decoding, which is defined as:

$$e_n = f_n - \hat{f}_n$$

$$\&, \hat{f}_n = e_n + \hat{f}_n$$

Interframe & Intraframe Coding

It is used for video compression. The intraframe coding is also known as I-frame.

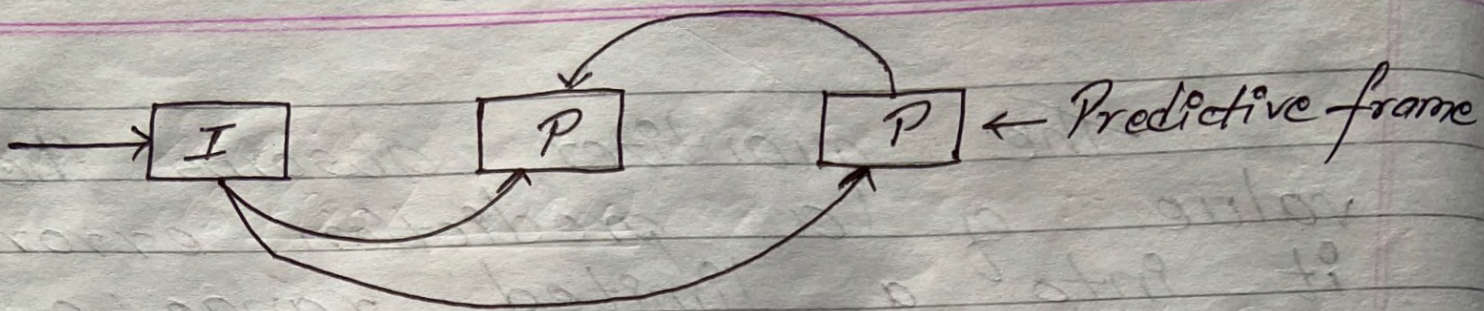
Intraframe compression uses only current frame for encoding.

Interframe coding is encoding and compression technique between the video signal.

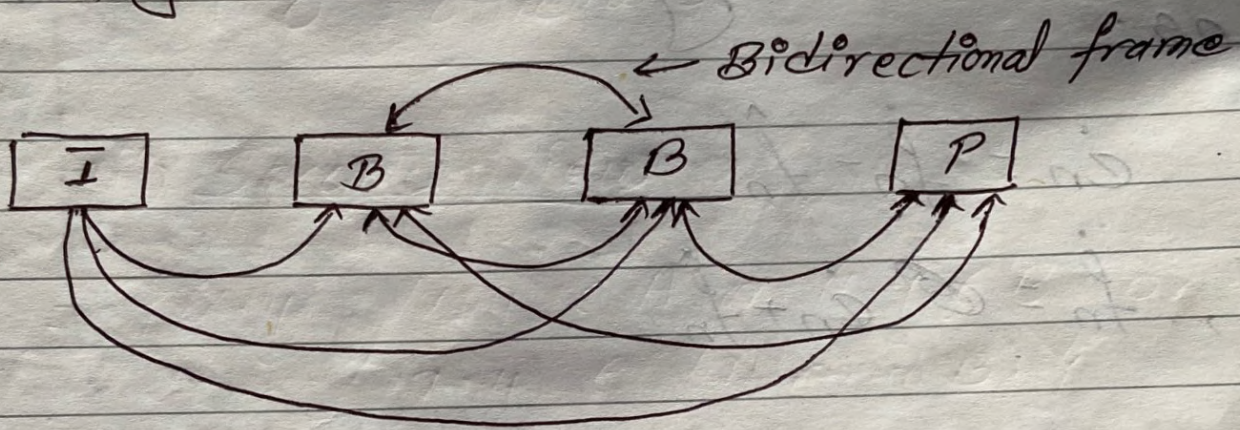
Interframe & Intraframe coding is categorized into 3 types, on the basis of predicted frame.

- (i) Predictive frame (P-frame)
- (ii) Intraframe or Independent frame (I-frame)
- (iii) Bidirectional frame (B-frame).

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(fig ca) : I and P frame)



(fig (b) : I, P and B frame)